

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 316 — Past Questions & Materials
by Renance

Course: MEE 316
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 6
Levels: 300

This study pack was compiled and restructured by Renance for MEE 316 students at Federal University of Petroleum Resources, Effurun. It brings together 6 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 6 — MEE 316 Thermodynamics and Heat Transfer

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MECHANICAL ENGINEERING

FIRST SEMESTER EXAMINATION - 2024/2025 ACADEMIC SESSION

COURSE CODE AND TITLE: MEE 316 -THERMODYNAMICS AND HEAT TRANSFER (2 UNITS)

INSTRUCTION: ANSWER ALL QUESTIONS

TIME ALLOTTED: 2 HOURS 45 MINUTES

QUESTION 1[12 MARKS]

(a) Imagine getting to your house on a sunny afternoon and your refrigerator, which should contain some chilled

drinks, has been OFF and refuses to start when switched ON. State, with a 1-sentence justification, which

amongst the condenser, evaporator, compressor, and expansion valve would be most likely to have failed.

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(b) A vessel in your organization of internship, designed for storing mixture of yuses, contains 3 kg of O₂, 5 kg of N₂,

and 12 kg of CH₄ at a time. What response would you give to your supervisor when asked to estimate the mass

fraction of each mixture component?

[4 marks]

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(c) The table hereunder specifies signs of entropy change, entropy transferred, and entropy generated in a generic closed system. An empty entry in the table connotes that the quantity can be positive, negative, or 0,

Classify as either possible or impossible, with justifications, the processes (i) to (iv). [6 marks]

Entropy Change Entropy Transferred Entropy Generated

(i) <0

(ii) <0

(iii) >0

(iv) <0

QUESTION 2[27 MARKS]

Congratulations! You've been accepted as an intern in a renowned power generation company in Nigeria, and your first

task is to appraise the design of a steam power plant working on an ideal Rankine cycle with superheating and -

reheating. The plant manager has briefed you that the plant has the following design specifications. Steam enters

the first-stage turbine at 10 MPa, 500°C, and expands to 5 MPa. It is then reheated to 450°C before entering the

second-stage turbine, where it expands to the condenser pressure of 0.1 MPa. The net power output is 100 MW.

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the specific net work done by the plant, (c) the specific total heat supplied to the plant (d) the thermal efficiency of

the cycle, (e) the mass flow rate of steam, (f) the actual rate of heat transfer Q_{out} from the condensing steam as it

passes through the condenser, in MW, and (g) the total exergy destroyed (irreversibility) in the power plant.

ii 1{Page

QUESTION 3[18 MARKS]

(a) The Centre for Thermofluids Studies, FUPRE, developed a robot, which is to work in the Thermofluids

laboratory. Considering the robot to be a solid suddenly immersed in a well stirred fluid. Carry out lumped

system analysis of the resultant heat transfer scenario.

(b) Derive the general heat conduction equation in cylindrical coordinates

QUESTION 4[17 MARKS]

(a) A truncated cone of height 10 cm has top and bottom diameters of 8 cm and

surface is stated to intercept 15 % of radiation leaving the top surface. Determine

the (i) top surface and conical side surface, and (ii) the side surface and itself.

(b) A plain wall 90 mm thick ($k = 0.18 \text{ W/m}^\circ\text{C}$) is insulated on one side while the other side is exposed to environment

at 80°C. The rate of heat generation within the wall is $1.3 \times 10^6 \text{ W/m}^3$. If the convective heat transfer

coefficient between the wall and the environment is $520 \text{ W/m}^2\text{}^\circ\text{C}$, determine the maximum temperature to

which the wall will be subjected.

d 16 cm respectively. The bottom

determine the shape factor between

Formulae sheet

Exergy balance equation - $\dot{E}_{\text{in}} - \dot{E}_{\text{out}} + \dot{Q}(1 - Z) = \dot{E}_{\text{D}}$

C

Specific physical exergy - $e_{pn} = (h - h_g) - T_{als} - S_a$

Extracted properties of saturated steam

P hiki/ke) he Ne Sr Se Srp V(m'/kg) V,

(MPa) (kJ/kg) (kJ/kg) _ (kJ/kg. K) (kJ/kg.K) (kJ/kg.K) (m'/kg)

0.1 417.5] 2675.1 2257.6 1.30273 7.3589 6.0562 1.04313 x 10° 1.6943

1 762.88 2777.7 2014.8. 2.13885 6.5859 4.4471 1,12720,x.10: na veut a1, 9438. worn

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10 1407.33 2724.5 1317.2 3.35918 5.6140 2.2548 1.45220 x 10° 0.018026 .

12 1490.79 2684.6 1193.8 3.49534 5.4922 1.9968 1.52596 x 10° 0.014263

Extracted properties of superheated steam

5 MPa 10MPa

T h S Vim' /kg) T h S V(m'/kg)

(°C) (ki/kg) (kJ/kg. K) (°C) _ (kJ/kg) (kJ/kg.K)

7 ty

450 3316.3 6.8187 63.30x10°

500 3433.9 6.9760 68.56 , 192 ~ ~ «500. 3374.0 6,597] 32.784 x 10°

600 3666.2 7.2586 _ 78.69.x 10° 600 3624.7 6.9022 _ 38.361 x ht

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Paper 2 of 6 — MEE 316 Thermodynamics and Heat transfer 2024/2025

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MECHANICAL ENGINEERING
FIRST SEMESTER EXAMINATION - 2024/2025 ACADEMIC SESSION

COURSE CODE AND TITLE: MEE 316 -THERMODYNAMICS AND HEAT TRANSFER (2 UNITS)

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[6 marks]

	Entropy Change	Entropy Transferred	Entropy Generated
(i)		<0	0
(ii)	<0	0	
(iii)	>0	0	<0
(iv)	<0		>0

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- (b) A plain wall 90 mm thick ($k = 0.18 \text{ W/}^\circ\text{C}$) is insulated on one side while the other side is exposed to environment at 80°C . The rate of heat generation within the wall is $1.3 \times 10^5 \text{ W/m}^3$. If the convective heat transfer coefficient between the wall and the environment is $520 \text{ W/m}^2\text{C}$, determine the maximum temperature to which the wall will be subjected.

Formulae sheet

Exergy balance equation -
$$\sum \dot{m}_i e_i + \dot{Q} \left(1 - \frac{T_a}{T_c}\right) = \sum \dot{m}_o e_o + \dot{W} + \dot{I}$$

Specific physical exergy - $e_{ph} = (h - h_a) - T_a(s - s_a)$

Extracted properties of saturated steam

P (MPa)	h_f (kJ/kg)	h_g (kJ/kg)	h_{fg} (kJ/kg)	S_f (kJ/kg.K)	S_g (kJ/kg.K)	S_{fg} (kJ/kg.K)	V_f (m ³ /kg)	V_g (m ³ /kg)
0.1	417.51	2675.1	2257.6	1.30273	7.3589	6.0562	1.04313×10^{-3}	1.6943
1	762.88	2777.7	2014.8	2.13885	6.5859	4.4471	1.12720×10^{-3}	0.19438
5	1154.22	2793.7	1639.5	2.92013	5.9725	3.0524	1.28515×10^{-3}	0.03944
10	1407.33	2724.5	1317.2	3.35918	5.6140	2.2548	1.45220×10^{-3}	0.018026
12	1490.79	2684.6	1193.8	3.49534	5.4922	1.9968	1.52596×10^{-3}	0.014263

Extracted properties of superheated steam

5 MPa

T (°C)	h (kJ/kg)	s (kJ/kg.K)	V(m ³ /kg)
385	3158.2	6.5895	56.10×10^{-3}
450	3316.3	6.8187	63.30×10^{-3}
500	3433.9	6.9760	68.56×10^{-3}
600	3666.2	7.2586	78.69×10^{-3}

10MPa

T (°C)	h (kJ/kg)	s (kJ/kg.K)	V(m ³ /kg)
450	3241.1	6.4194	29.752×10^{-3}
500	3374.0	6.5971	32.784×10^{-3}
600	3624.7	6.9022	38.361×10^{-3}

② $\frac{T - T_o}{T_i - T_o} = e^{\frac{hA(T_o - T_i)}{PVC}}$

$T = T_{\text{fluid}}$
 $T_o = T_{\text{ambient}}$
 $T_i = \text{Surface}$
 $P = \text{Perimeter}$
 $V = \text{Volume}$
 $h = \text{Heat transfer coefficient}$
 $A = \text{Area of surface}$
 $PVC = \text{Product of volume and convective coefficient}$

Paper 3 of 6 — MEE 316 Thermodynamics and Heat Transfer 2024/2025.

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MECHANICAL ENGINEERING

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COURSE CODE AND TITLE: MEE 316 -THERMODYNAMICS AND HEAT TRANSFER (2 UNITS)

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(iii) >0 0 - <0

|(iv) | <0 >0

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Formulae sheet

Exergy balance equation - $\dot{E}_x = \dot{Q}(1 - T_0/T) - \dot{W} + \dot{E}_x$

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Extracted properties of saturated steam

P (MPa) h_f (kJ/kg) h_g (kJ/kg) s_f (kJ/kg.K) s_g (kJ/kg.K) v_f (m³/kg) v_g (m³/kg)

(MPa) (kJ/kg) (kJ/kg) (kJ/kg.K) (kJ/kg.K) : (m³/kg)

0.1 4) 7.51 2675.1 2257.6 ~ 1330273 7.3589 6.0562 1.04313×10^{-6} 1.6943 -

1 762.88 2977.7 2014.8 2113885 6.5859 4.4471 1.12720×10^{-6} 0.19438

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2 1490.79 2684.6 1193.8 3.49534 5.4922 1.9968 1.52596×10^{-6} 0.014263

Extracted properties of superheated steam

5 MPa 10 MPa

T (K) s (kJ/kg.K) v (m³/kg) T (K) s (kJ/kg.K) v (m³/kg)

(K) (kJ/kg.K) v (m³/kg) T (K) s (kJ/kg.K) v (m³/kg)

35 315.2 6580.5 0.00101 35 315.2 6580.5 0.00101

450 331.63 6.8187 6.230×10^{-4} 450 324.11 6.4194 2.9752×10^{-4}

500 343.39 6.9760 6.856×10^{-4} 500 337.40 6.5971 3.2784×10^{-4}

600 366.2 7.2586 7.869×10^{-4} 600 362.47 6.9022 3.8361×10^{-4}

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Paper 4 of 6 — MEE 316 THERMODYNAMICS & HEAT TRANSFER 2024

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

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Formulae sheet

Exergy balance equation - $\dot{Q} = \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}} = \dot{Q}_{\text{net}}$ Me; + Q (1 = =| = ». Melo KW+T -

Cc?

Specific physical exergy - $e_y = (h - h_g) - T_a(S - S_a)$

Extracted properties of saturated steam

P hikikg) h, hy Ss A Ste Viamkg) Vv:

(MPa) (kJ/kg) (kJ/kg) (kJ/kg.K) _(/kg.K) _(/kg.K) (m°/kg)

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ieeee! s Viney T bh S Vim'ikg)

(°C) (kJ/kg) _(/kg.K) Jb. (C) (kJ/kg) (kJ/kg.K) ;

3158.2 6.5895 56.10 x 10 ~

oe 3316.3 68187 \$3.30 x 10° AS) 3241.1 6.4194 29.752 x 10°

500 3433.9 6.9760 «© 68.56x 10%. SW) 3374.0 6.5971 32.784 x 10°

500 3666.2 7.2586 -78.69x10" _—61X)_—3624.7 _—6.9022—38.361 x 10"

Paper 5 of 6 — MEE 316 - Thermodynamics and Heat Transfer

Level 300 • First Semester • Handout

Federal University of Petroleum Resources, Effurun,
Nigeria

Mechanical Engineering Department, College of Engineering and Technology

Course: Thermodynamics and Heat Transfer – MEE 316
(Thermodynamics Part)

Course Lecturer:

Joseph OYEKALE (PhD-Doctor Europaeus, MNSE, R.Engr)

2024 - 2025 Academic Session

1

Outline

✓ Thermodynamics

- General introduction and thermodynamic working fluids (mixtures)
- Vapour power cycles (Carnot and ideal Rankine cycles)
- Rankine cycle with superheat and reheat
- Second law of thermodynamics and introduction to exergy

✓ Basic Thermodynamics Cycle in Python

- Computer applications for solving thermodynamics cycle problems!
- Practical case study – course project!

2

Learning Outcomes

At the end of the Thermodynamics part of MEE 316 you (students) should be able to:

- > state the basic components of a vapour power plant operating on Rankine cycle and explain the various thermodynamic processes involved;
- > compute entropy generation and exergy rates in vapour power cycles and other systems, and discuss the implications of entropy generation/exergy destruction on the performance of thermodynamic systems;
- > solve basic thermodynamics cycle application problems analytically and numerically using dedicated computer programs in Python.

3

Organization

Flipped classroom and peer instruction teaching/learning approach would be adopted.

That implies that students should STUDY the notes in this document carefully before classes and consult textbooks and Generative AI Tools (GAITs) to learn more about the relevant topics.

Lecture periods will be for questioning on vital concepts and application case studies (examples)!

Some good textbooks consulted while preparing this and subsequent lecture slides (intellectual credit to authors of these texts):

- ✓ Thermodynamics: An Engineering Approach, 5th edition by Yunus A. Çengel and Michael A. Boles.
- ✓ Energy Systems: A New Approach to Engineering Thermodynamics by Renaud Gicquel
- ✓ Çengel, Y. A. Heat and Mass Transfer: A Practical Approach.
- ✓ Engineering Heat and Mass Transfer by Mahesh M. Rathore. 4

General Introduction and Thermodynamic Working Fluid

How does a steam (vapour) power plant work? What basic components and thermodynamic processes are involved?

5

Mixtures As Working Fluids

➤ Rules for determining non-reacting gas mixtures

There are several cases where thermodynamic systems use mixtures, instead of pure substances, as working fluid. Analyses of such systems require adequate understanding of the mixture properties.

Although it is possible to compile properties of some standard mixtures such as air and make them readily available in tables, it would be arduous, if not practically impossible, to capture all kinds of mixtures in such tables.

Thus, it is imperative to set rules for determining properties of mixtures in the study of thermodynamic systems, which is the specific goal of this topic.

To determine the properties of a mixture, both its composition and properties of each of the components must be known.

There are basically two ways of describing compositions of a mixture: (i) by molar analysis (description based on the number of moles of each component) and (ii) by gravimetric analysis (description based on the mass of each component). 6

Mixtures As Working Fluids

➤ Rules for determining nonreacting gas mixtures

Some general rules of describing compositions of mixtures are:

- The mass of a mixture is equal to the sum of the masses of its components;
- The number of moles of a nonreacting gas is equal to the sum of the number of moles of its components;

Mathematically

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Mixtures As Working Fluids

➤ Rules for determining nonreacting gas mixtures

Some general rules of describing compositions of mixtures are:

- The ratio of number of moles of a component to the number of moles of a mixture is known as the mole fraction (y);
- The ratio of mass of a component to the mass of mixture is called the mass fraction (mf) of such component;
- The sum of the mole fractions of a mixture is equal to 1, and that of mass fractions of a mixture is as well equal to 1.

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Mixtures As Working Fluids

- > Some other vital principles for analysing mixture properties
- The apparent (or average) molar mass and the gas rate constant of a mixture can be expressed as:
- The molar mass of a mixture can also be expressed as:
- Mass and mole fractions of a mixture are related by:

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Mixtures As Working Fluids

> Example 1

Consider a gas mixture that consists of 3 kg of pO_2 , 5 kg of yO_2 , and 12 kg of O_2 , as shown in the figure below. Determine (a) the mass fraction of each component, (b) the mole fraction of each component, and (c) the average molar mass and gas constant of the mixture.

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Mixtures As Working Fluids

> P-V-T Behaviour of Gas Mixtures

When it can be assumed that the molecule of a gas is far apart from other molecules, such that its behaviour is not influenced by the presence of the surrounding molecules, the gas is said to be ideal.

The P-v-T behavior of an ideal gas is expressed by the simple relation $Pv=RT$, which is called the ideal-gas equation of state. The P-v-T behavior of real gases is expressed by more complex equations of state or by $Pv=ZRT$, where Z is the compressibility factor.

The prediction of the P-v-T behavior of gas mixtures is usually based on two models:

- Dalton's law of additive pressures,
- Amagat's law of additive volumes.

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Mixtures As Working Fluids

> P-V-T Behaviour of Gas Mixtures

Dalton's law of additive pressures states that the pressure of a gas mixture is equal to the sum of the pressures each gas would exert if it existed alone at the mixture temperature and volume.

Amagat's law of additive volumes states that the volume of a gas mixture is equal to the sum of the volumes each gas would occupy if it existed alone at the mixture temperature and pressure.

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Mixtures As Working Fluids

> Ideal Gas Mixtures

For ideal gases, y_i and ϕ_i can be related to ϕ by using the ideal-gas relation for both the components and the gas mixture:

Therefore,

The quantity $\phi_i y_i$ is called the partial pressure (identical to the component pressure for ideal gases), and the quantity $\phi_i \phi$ is called the partial volume (identical to the component volume for ideal gases).

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Mixtures As Working Fluids

> Ideal Gas Mixtures

The composition of an ideal-gas mixture (such as the exhaust gases leaving a combustion chamber) is frequently determined by a volumetric analysis (called the Orsat Analysis) and equation (**) above.

Please check out how these principles are adapted to suit analysis of real gas mixtures!

> Exercise

A rigid tank contains 2 kmol of y_i and 6 kmol of ϕ_i gases at 300 K and 15 MPa. Estimate the number of mole in the tank and the volume of the tank on the basis of the ideal-gas equation of state.

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Vapour Power Cycle

> Background

As explained last week, power cycles involve phase change of working fluid. One key example is a case where water (steam) is the working fluid, as in steam power cycle. To allow for the use of other liquids other than water, they are generally called vapour power cycles.

Carnot Vapor Cycle

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Vapour Power Cycle

> The Carnot vapour cycle

The heat engine may be composed of the following components.

The working fluid, steam (water), undergoes a thermodynamic cycle from 1-2-3-4-1. The cycle is shown on the following T-s diagram.

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Vapour Power Cycle

➤ The Carnot vapour cycle Carnot Vapor Cycle Using Steam

700

600

500

6000 kPa

400

T [C]

300 2 100 kPa

3

200

100 1 4

0

0.0 1.0 2.0 3.0 4.0 5.0 6.0 7.0 8.0 9.0 10.0

s [kJ/kg-K]

The thermal efficiency of this cycle is given as: $\eta_{th, Carnot} = 1 - \frac{T_{out}}{T_{in}}$

$\eta_{th, Carnot} = 1 - \frac{T_{out}}{T_{in}}$

$Q_{in} - Q_{out}$

T_L

$= 1 - \frac{T_L}{T_H}$

T_H

Note the effect of T_H and T_L on $\eta_{th, Carnot}$.

- The larger the T_H the larger the $\eta_{th, Carnot}$
- The smaller the T_L the larger the $\eta_{th, Carnot}$

Vapour Power Cycle

➤ The Carnot vapour cycle

To increase the thermal efficiency in any power cycle, we try to increase the maximum temperature at which heat is added.

Reasons why the Carnot cycle is not used:

- Pumping process 1-2 requires the pumping of a mixture of saturated liquid and saturated vapor at state 1 and the delivery of a saturated liquid at state 2.
- To superheat the steam to take advantage of a higher temperature, elaborate controls are required to keep T_H constant while the steam expands and does work.

To resolve the difficulties associated with the Carnot cycle, the Rankine cycle was introduced.

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Vapour Power Cycle

➤ The Rankine cycle

The simple Rankine cycle has the same component layout as the Carnot cycle shown

above. The simple Rankine cycle continues the condensation process 4-1 until the saturated liquid line is reached.

Ideal Rankine Cycle Processes

Process Description

1-2 Isentropic compression in pump

2-3 Constant pressure heat addition in boiler

3-4 Isentropic expansion in turbine

4-1 Constant pressure heat rejection in condenser

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Vapour Power Cycle

> The Rankine cycle

The T-s diagram for the Rankine cycle is given below. Locate the processes for heat transfer and work on the diagram.

Rankine Vapor Power Cycle

500

6000 kPa

400

3

300

T [C] 200

10 kPa

100 2

4

1

0

0 2 4 6 8 10 12

s [kJ/kg-K]

Examples:

1) A heat engine is supplied heat at the rate of 1700 kJ/min and gives an output of 9 kW. Determine the thermal efficiency and the rate of heat rejection. 20

Vapour Power Cycle

> The Rankine cycle

Examples:

2) Consider a steam power plant operating on the simple ideal Rankine cycle. Steam leaves the boiler as superheated vapour at 6 Mpa, 350 oC, and is condensed at 10 kPa. Determine the specific work input into the pump, specific work done by the turbine, specific heat input into the plant and specific heat rejected. What is the thermal efficiency of this cycle?

In a Rankine cycle, the steam at inlet to turbine is saturated at a pressure of 30

bar and the exhaust pressure is 0.25 bar. Determine : (i) The pump work (ii) Turbine work (iii) Rankine efficiency (iv) Condenser heat flow (v) Dryness at the end of expansion. Assume flow rate of 10 kg/s.

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Vapour Power Cycle

> The Rankine cycle – Ideal vs Real

Although our discussions on the Rankine cycle concern mainly the simple/ideal cycle, it is necessary to highlight the main features that differentiate it from the actual Rankine cycle, as illustrated in the figures below.

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Vapour Power Cycle

> The ideal Rankine cycle – ways to improve cycle efficiency

- Superheat the vapor

Average temperature is higher during heat addition.

Moisture is reduced at turbine exit (we want $x_4 > 0.85$).

- Increase boiler pressure (for fixed maximum temperature)

Availability of steam is higher at higher pressures.

Moisture is increased at turbine exit.

- Lower condenser pressure

Less energy is lost to surroundings.

Moisture is increased at turbine exit.

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Vapour Power Cycle

> The Reheat Cycle

As the boiler pressure is increased in the simple Rankine cycle, not only does the thermal efficiency increase, but also the turbine exit moisture increases. The reheat cycle allows the use of higher boiler pressures and provides a means to keep the turbine exit moisture ($x > 0.85$ to 0.90) at an acceptable level.

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Vapour Power Cycle

> The Reheat Cycle

Main Processes

Component Process Energy Balance (First Law Result)

Boiler Const. $P \dot{q}_{in} = (h_3 - h_2) + (h_5 - h_4)$

Turbine Isentropic $w_{out} = (h_3 - h_4) + (h_5 - h_6)$

Condenser Const. $P \dot{q}_{out} = (h_6 - h_1)$

Pump Isentropic $w_{in} = (h_2 - h_1) = v_1(P_2 - P_1)$

> Example

Consider a steam power plant operating on the ideal reheat Rankine cycle. Steam enters the high-pressure

turbine at 15 MPa and 600°C and is condensed in the condenser at a pressure of 10 kPa. If the steam is

reheated at 4 Mpa to the inlet temperature of the high-pressure turbine, determine (a) the total heat input

into the cycle, (b) the heat rejected, (c) the net work done and (d) the thermal efficiency of the cycle.

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THANK YOU FOR YOUR ATTENTION!

Any other questions?

Paper 6 of 6 — MEE 316 - Thermodynamics and Heat Transfer

Level 300 • First Semester • Handout

Federal University of Petroleum Resources, Effurun,
Nigeria

Mechanical Engineering Department, College of Engineering and Technology

Course: Thermodynamics and Heat Transfer – MEE 316

Thermodynamics Part

Course Lecturer:

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2024 - 2025 Academic Session

1

The Second Law of Thermodynamics

> Background

- Why does water never run up the waterfall?
- The second law defines entropy (S) in a system.
- Statements of the second law:

> Kelvin Planck's statement

✓ It is impossible for any device that operates on a cycle to receive heat from a single reservoir and produce a net amount of work.

✓ The Kelvin-Planck statement of the second law of thermodynamics states that no heat engine can produce a net amount of work while exchanging heat with a single reservoir only. In other words, the maximum possible efficiency is less than 100 percent.

> Clausius' statement

✓ The Clausius statement of the second law states that it is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a lower-temperature body to a higher-temperature body.

The Second Law of Thermodynamics

> Background

• Other Statements:

✓ It is impossible for any device that operates on a cycle to receive heat from a single reservoir and produce a net amount of work.

✓ Heat can flow spontaneously from a hot object to a cold object; it will not flow spontaneously from a cold object to a hot object.

> Irreversibilities in Thermodynamic Systems

Unlike in reversible processes where state properties could be restored perfectly to their original state even after the onset of a process, no state properties could be returned to their exact initial positions in irreversible processes.

Thermodynamic reversibility is ideal; all real processes are irreversible!

For closed and isolated systems, 2nd Law of Thermodynamics is defined mathematically as:

◆□ā

$\dot{y}_2 - \dot{y}_1 = \dot{y}_{\text{in}} - \dot{y}_{\text{out}} + \dot{y}_{\text{gen}}$ where $\dot{y}_{\text{gen}} \geq 0$

For reversible systems, $\dot{y}_{\text{gen}} = 0$

For irreversible systems, $\dot{y}_{\text{gen}} > 0$

The Second Law of Thermodynamics

➤ Basic application steps of the 2nd Law

1. Define the system
 2. Identify initial and final states
 3. Identify heat transfer interactions and irreversibilities
 4. Write the 2nd Law
 5. Expand the LHS to get a useful expression for the change in entropy stored
 6. Solve for the unknown parameters
- If there's irreversibility the 2nd Law is usually used
 - If reversible then change in entropy stored = transferred entropy

Please use Kelvin not degree C for entropy!

The Second Law of Thermodynamics

➤ Irreversibilities in Thermodynamic Systems

- Some causes of irreversibilities:
 - ✓ Friction
 - ✓ Unrestrained expansion of gases
 - ✓ Heat transfer through a finite temperature difference
 - ✓ Mixing of two different substances, etc

➤ Some Applications of the Second Law

- Heat Engines (Kelvin Planck's Statement; treated earlier in Lecture 2). Most power generation plants are examples of heat engines, regardless of heat source.
- Refrigerators, Air Conditioners and Heat Pumps (Clausius' Statement; they work in the reverse order of heat engines).

The Second Law of Thermodynamics

➤ Applications based on the Clausius' statement

- ✓ Refrigerators, Air conditioners and Heat Pumps

By receiving external work, heat is extracted from the cold reservoir and exhausted to the hot reservoir.

R&A performance is measured by the

coefficient of performance (COP):

$$\dot{Q}_H / \dot{Q}_C$$

$$T_H / (T_H - T_C) =$$

$$\dot{Q}_H / \dot{Q}_C = 2 \dot{Q}_C / \dot{Q}_C$$

Ideal COP could be obtained from just the temperatures, as:

$$T_H /$$

$$T_H - T_C =$$

$$T_H / 2 T_H$$

Refrigerator and Air Conditioner (R&A)

The Second Law of Thermodynamics

> Applications based on the Clausius' statement

✓ Refrigerators, Air conditioners and Heat Pumps

A heat pump can heat a house in the harmattan.

Heat pump is also measured by the coefficient of performance (COP):

$$\dot{Q}_H / \dot{Q}_C$$

$$T_H / (T_H - T_C) =$$

$$\dot{Q}_H / \dot{Q}_C = 2 \dot{Q}_C / \dot{Q}_C$$

Heat Pumps

Entropy and the Second Law of Thermodynamics

> Other facts about entropy:

The total entropy of an isolated system never decreases.

Natural processes tend to move toward a state of greater disorder.

Thermal equilibrium is a similar process4the uniform final state has more disorder than the separate temperatures in the initial state.

The higher the entropy in a thermodynamic system, the higher the irreversibility, and the lower the efficiency.

So, what would be your goal in a thermodynamic system design, to increase entropy or minimize it?

Entropy and the Second Law of Thermodynamics

> Some concept questions (Intellectual credit to MIT F22 2.005 Tutors)

i) Can the entropy change of a closed system be zero during an irreversible process?

Explain.

ii) <Strength of irreversible processes=. A heat transfer of 2000 kJ occurs from a large reservoir at 1000 K to another large reservoir at either a) 200 K or b) 800 K (see Figure below). The reservoirs maintain constant temperature. The rod connecting the two reservoirs can be considered massless. Determine which heat transfer process is <more irreversible= (Hint: which of case a) or case b) will lead to the larger entropy generation?). Where is the

entropy generated?

Entropy and the Second Law of Thermodynamics

> Some concept questions (Intellectual credit to MIT F22 2.005 Tutors)

iii) Based on the sign of the following three quantities, classify the following processes a)-f) for a closed system as either possible or impossible, and very briefly mention why. The empty entries in the table simply mean that the quantity can be 0, > 0 or < 0 (i.e. it is not specified).

Exergy and the Second Law of Thermodynamics

> Anergy vs Exergy

One other important consequence of the second law is that, in any natural process, some energy becomes unavailable to do useful work.

This unavailable energy is called anergy, while the maximum available energy to do useful work is known as exergy.

It is possible to obtain exergy in all the states of a thermodynamic cycle.

Depending on the nature of thermodynamic system being analysed, specific exergy at each state could contain four components: the physical exergy, the chemical exergy, the kinetic exergy and the potential exergy.

In most cases, the potential and kinetic exergy components cancel out, and if no significant chemical reactions (such as combustion/mixing) take place, it may be sufficient to employ just the physical exergy to define thermodynamic states and thus, the system.

The physical exergy can be obtained from: $\dot{E}_{ph} = \dot{m} (h - h_0 - T_0 (s - s_0))$, where properties with subscript 0 are for the environment.

Exergy and the Second Law of Thermodynamics

> Exergy balance and assessment of thermodynamic components

In addition to conservation of mass and energy that should be ensured in the design of thermodynamic systems, exergy also should be conserved, if the system would satisfy the second law.

The three of them constitute the fundamental balance equations in thermodynamics, defined as:

$$\dot{m}_i \dot{y}_i = \dot{m}_e \dot{y}_e + \dot{A}$$

Subscript i stands for inlet and e for exit. $\dot{A}$ is the

$$\dot{m}_i h_i + \dot{Q} = \dot{m}_e h_e + \dot{W}$$

temperature at the component boundary. $\dot{I}$ in the third

$\dot{A}$ equation denotes irreversibility, as discussed.

$$\dot{m}_i e_i + \dot{Q} = \dot{m}_e e_e + \dot{W} + \dot{I}$$

$\dot{A}$

Let's now see how this can be applied.

Exergy and the Second Law of Thermodynamics

> Examples

1) A 3 bedroom bungalow being constructed in Ugbomro area of Delta State requires 2

$\times 10^8$ J/h for heating during harmattan. Heat pump is used to absorb thermal energy from cold air outside and to send heat inside the house. The work required to operate the heat pump is 3×10^7 J/h. Determine: (i) Heat abstracted from outside in kW; (ii) the coefficient of performance of the heat pump.

2) Consider a steam power plant working on ideal Rankine cycle with superheat and reheat. Steam enters the first-stage turbine at 10 MPa, 500 °C, and expands to 5 MPa. It is then reheated to 450 °C before entering the second-stage turbine, where it expands to the condenser pressure of 0.1 MPa. The net power output is 100 MW. Assume perfect thermal insulation for the pump and expander and take the average surface temperature as 1090 °C for the boiler and the reheater and 50 °C for the condenser. The specific enthalpy and specific entropy of the surrounding air, obtained at 17 °C and 0.1 MPa, are given as 71.355 kJ/kg and 0.2533 kJ/kg.K, respectively. Solve for: (a) the temperature of steam at the re-heater inlet, (b) the specific net work done by the plant, (c) the specific total heat supplied to the plant, (d) the thermal efficiency of the cycle, (e) the mass flow rate of steam (f) the actual rate of heat transfer Q_{out} from the condensing steam as it passes through the condenser, in MW, and (g) the total exergy destroyed in the power plant.

THANK YOU FOR YOUR ATTENTION!

Any other questions?